Project 1: Network Scanning & Information Gathering

Aim:

To perform network scanning using tools like Nmap or Zenmap to discover live hosts, open ports, and running services.

Tools Used:

- Nmap
- Zenmap
- Command Prompt

Theory:

Network scanning is a technique used to discover devices on a network and identify open ports and services. It helps in detecting vulnerabilities and understanding network security. Nmap is a popular tool used for scanning networks and gathering information.

Commands Used:

- nmap -sn 192.168.1.0/24
- nmap -sS 192.168.1.99
- nmap -A 192.168.1.99

Procedure:

1. Installed Nmap and Zenmap on the system.
2. Opened Command Prompt and found the local IP address using the ipconfig command.
3. Identified the network range as 192.168.1.0/24.
4. Performed host discovery using the command nmap -sn 192.168.1.0/24.
5. Scanned the target system (192.168.1.99) using SYN scan to identify open ports.
6. Performed an aggressive scan using nmap -A to detect operating system and running services.
7. Observed and recorded the output results.

Screenshots:

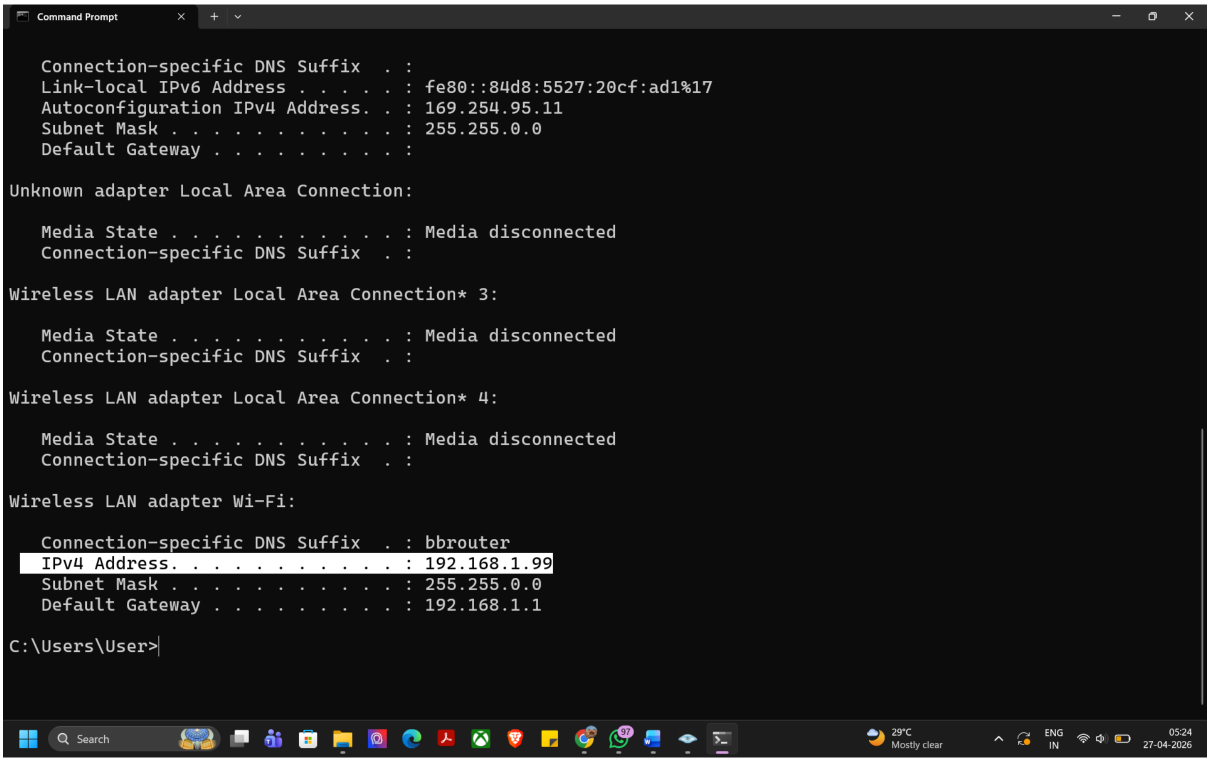

Figure 1 Display of local IP address using ipconfig command

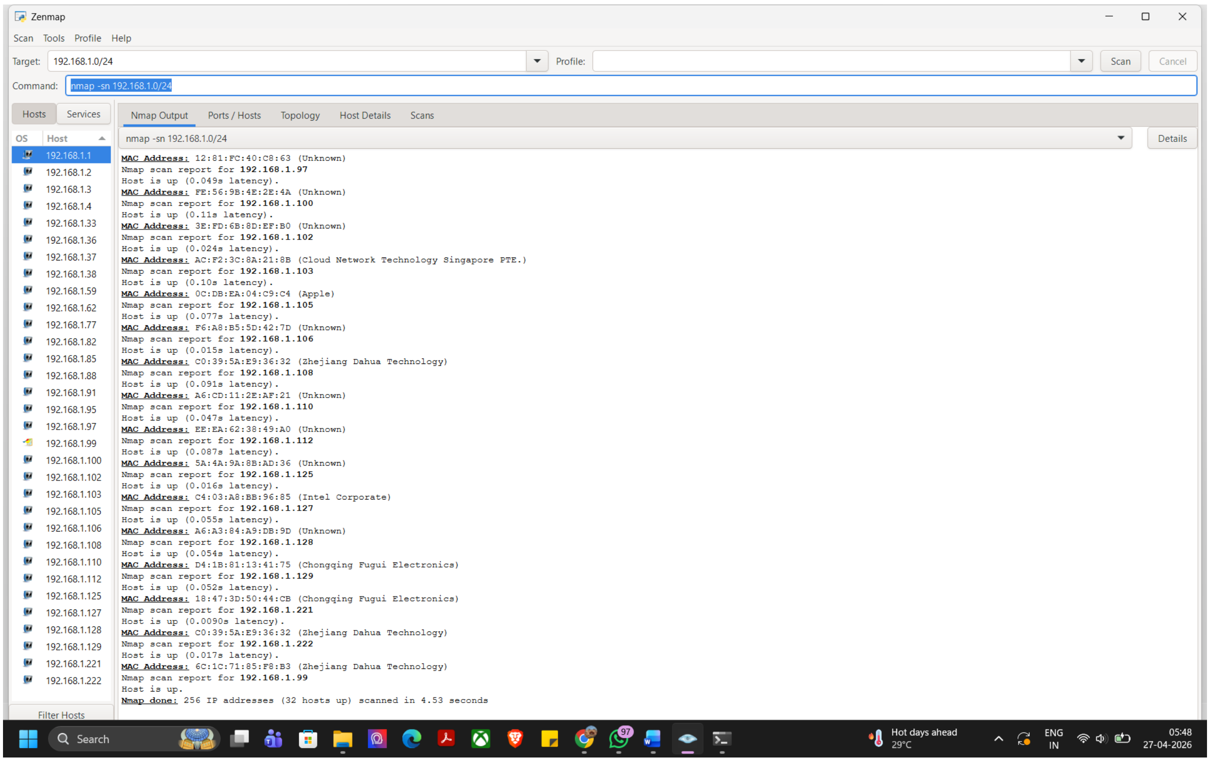

Figure 2 Network host discovery using Nmap showing active devices in the local network (192.168.1.0/24)

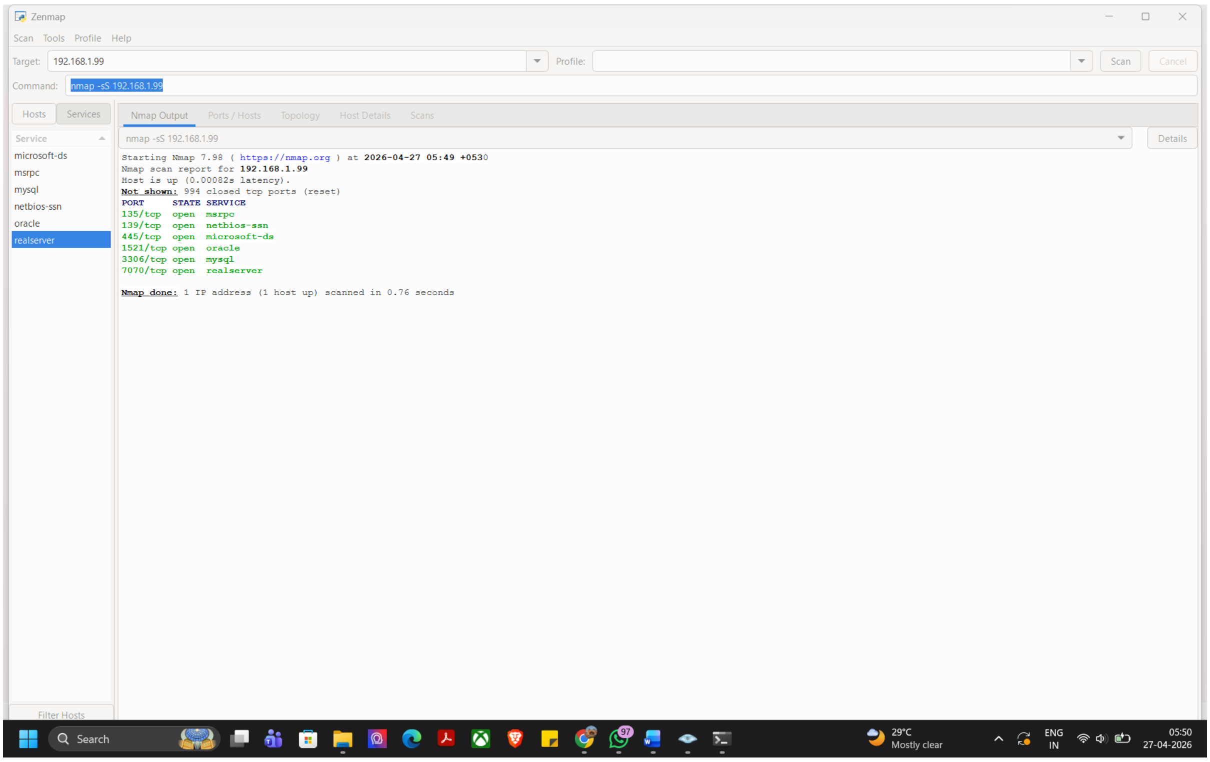

Figure 3 SYN scan performed using Nmap displaying open ports and their states on target system (192.168.1.99)

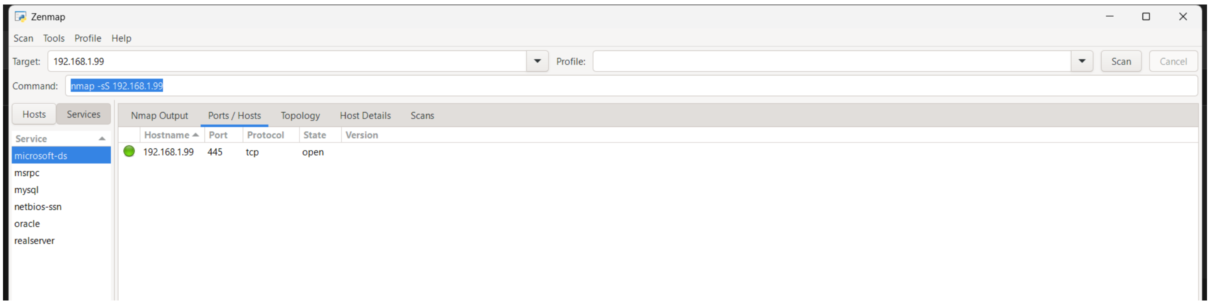

Figure 4 SYN scan using Nmap showing open port 445 (SMB service) on target system (192.168.1.99)

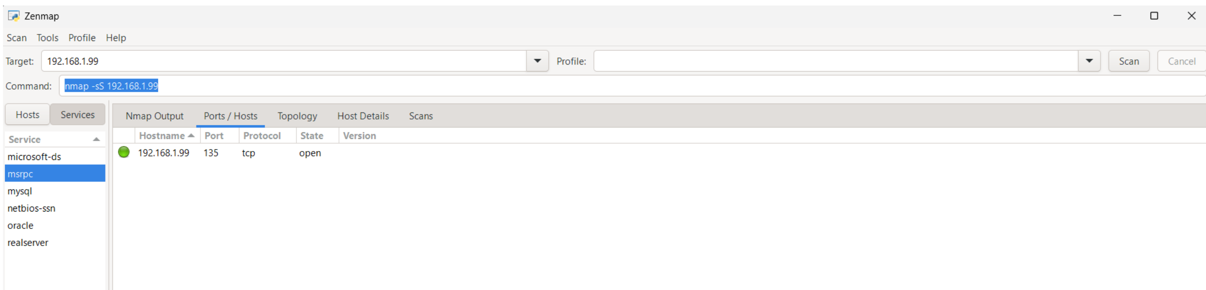

Figure 5 SYN scan using Nmap displaying open port 135 (RPC service) on target system (192.168.1.99)

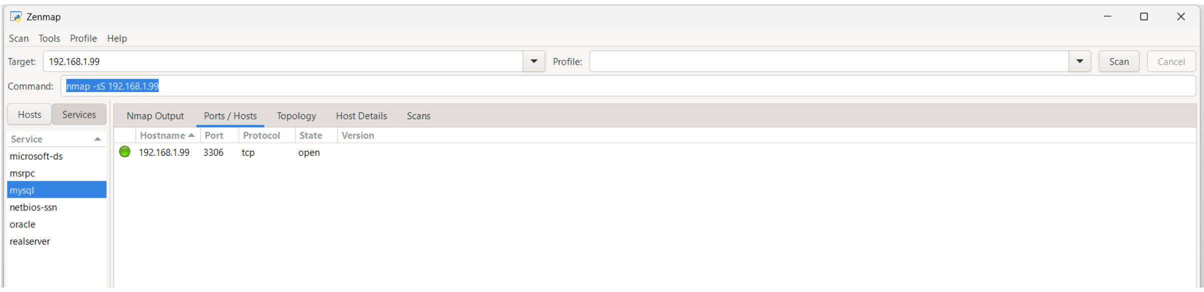

Figure 6 SYN scan using Nmap showing open port 3306 (MySQL service) on the target system (192.168.1.99)

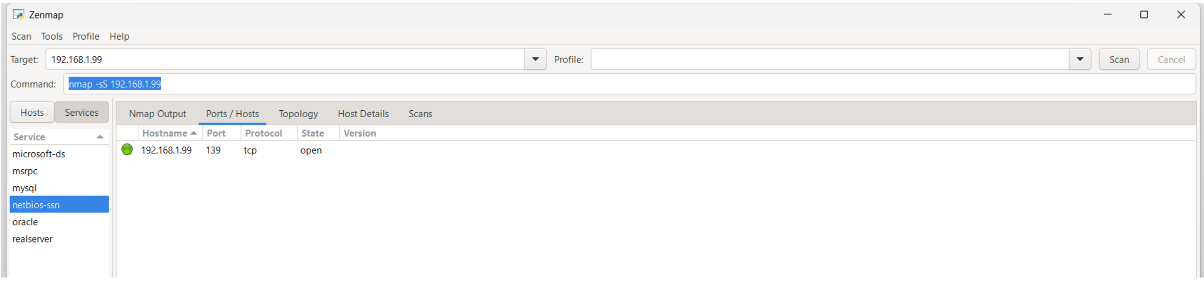

Figure 7 SYN scan using Nmap displaying open port 139 (NetBIOS service) on the target system (192.168.1.99)

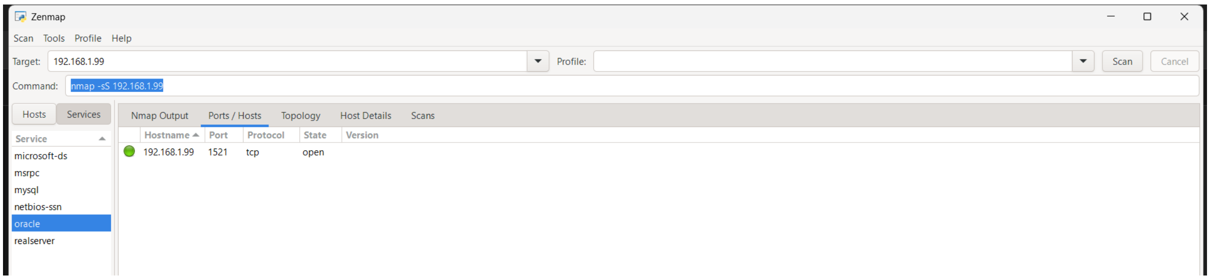

Figure 8 SYN scan using Nmap showing open port 1521 (Oracle database service) on the target system (192.168.1.99)

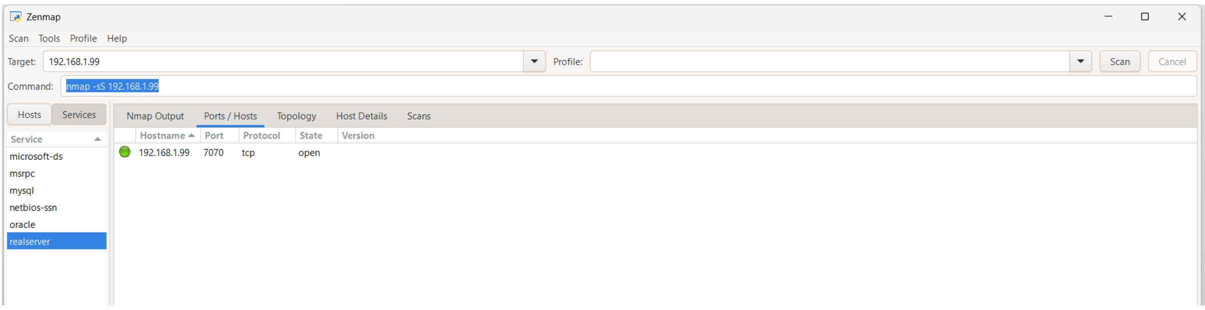

Figure 9 SYN scan using Nmap showing open port 7070 on the target system (192.168.1.99)

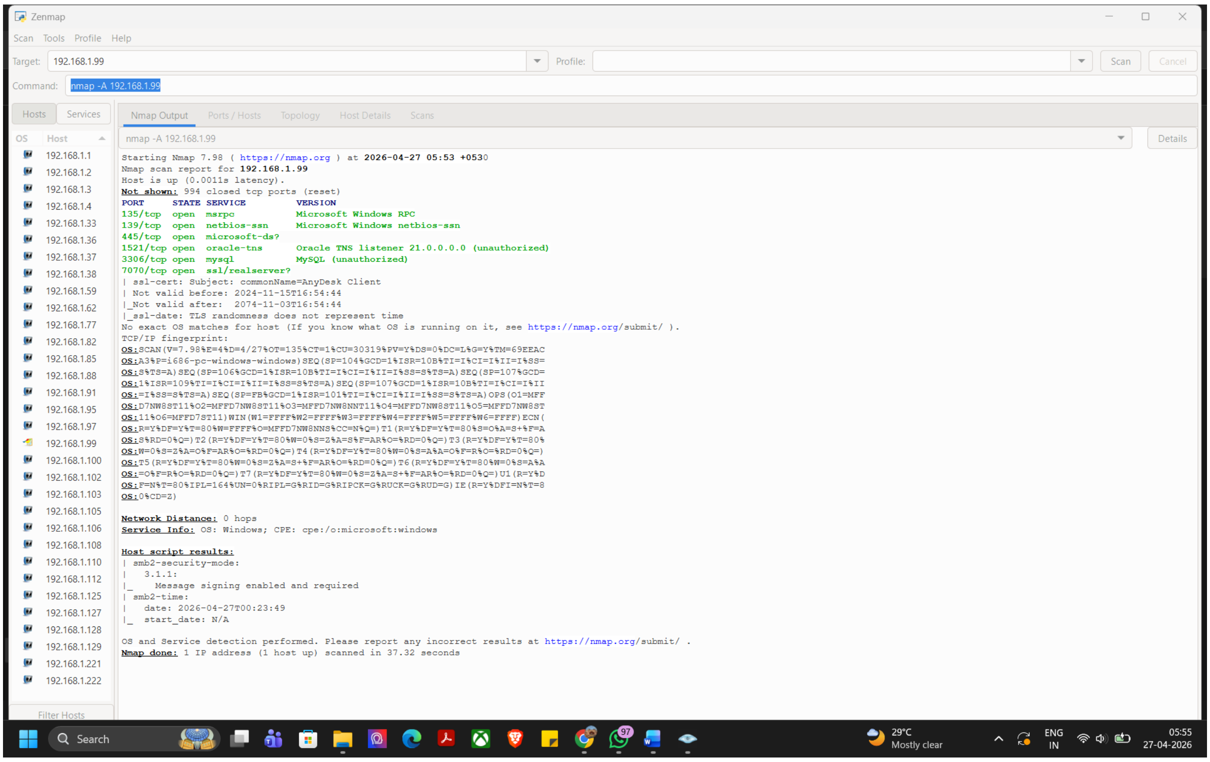

Figure 10 Aggressive scan using Nmap showing detailed information including services and operating system of the target system

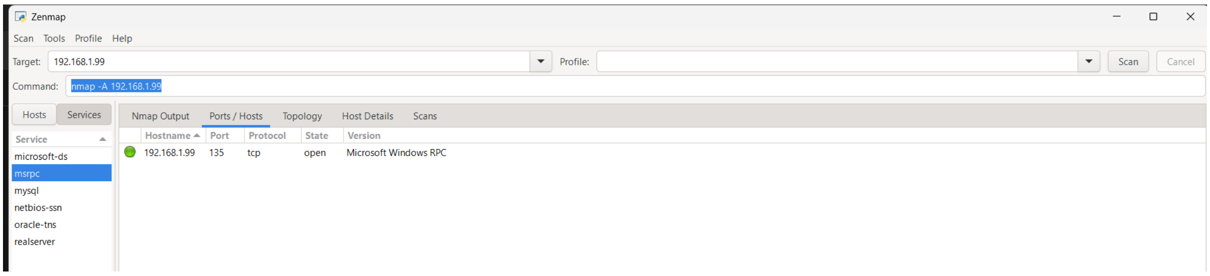

Figure 11 Aggressive scan using Nmap showing open port 135 (Microsoft Windows RPC service) with service details on the target system (192.168.1.99)

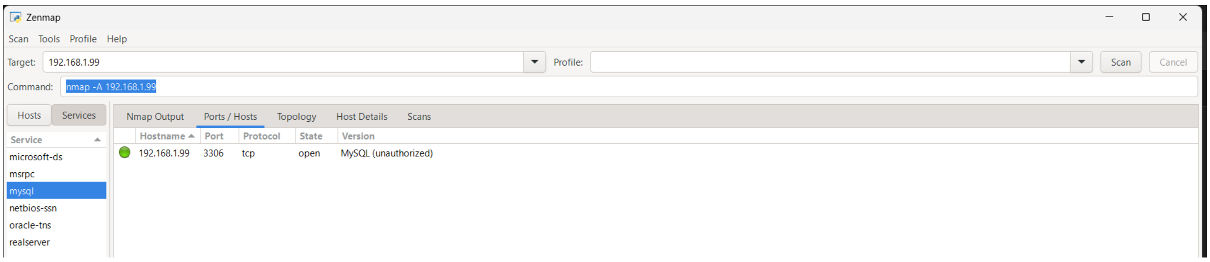

Figure 12 Aggressive scan using Nmap displaying open port 3306 (MySQL service) with version information on the target system (192.168.1.99)

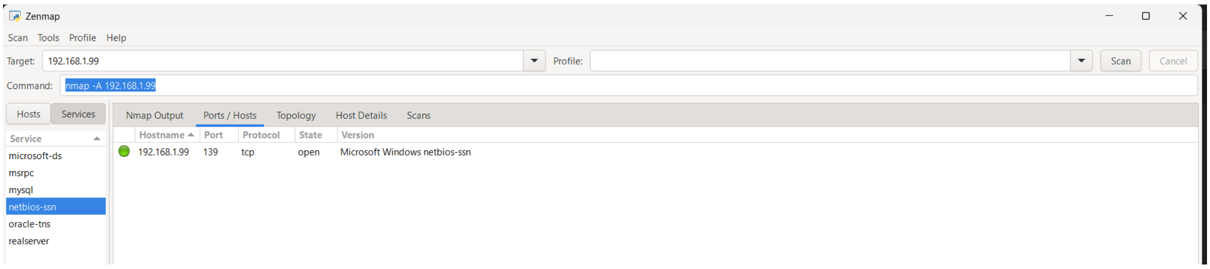

Figure 13 Aggressive scan using Nmap showing open port 139 (Microsoft Windows NetBIOS-SSN service) with version details on the target system (192.168.1.99)

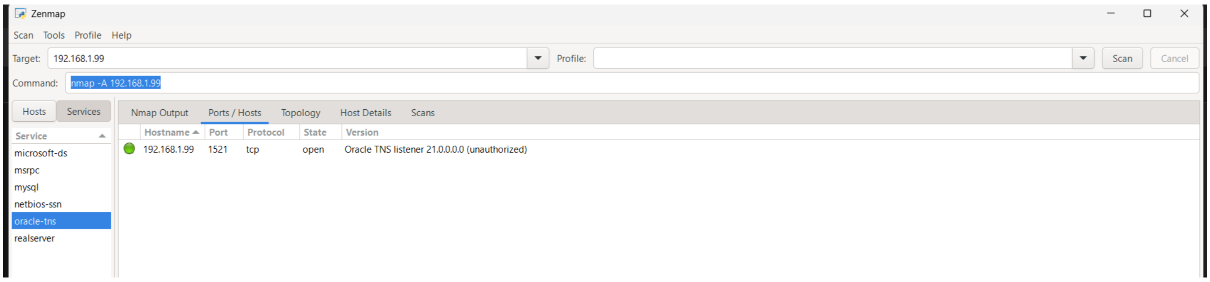

Figure 14 Aggressive scan using Nmap displaying open port 1521 (Oracle TNS Listener service) with version information on the target system (192.168.1.99)

Observations:

1. The network scan detected active hosts in the local network.
2. The target system with IP address 192.168.1.99 was found to be active.
3. Open ports such as 135 (RPC), 139 (NetBIOS), 445 (SMB), 3306 (MySQL), 1521 (Oracle), and 7070 were identified.
4. Services running on open ports were displayed.
5. The aggressive scan provided information about the operating system and service versions.

Result:

The network scanning process was successfully performed using Nmap. Live hosts, open ports, and running services were identified for the target system.

Conclusion:

Network scanning is an important step in cybersecurity for identifying active devices and potential vulnerabilities. Using tools like Nmap helps in understanding system exposure and improving network security. It also helps administrators take preventive measures against potential attacks.

Ethical Statement:

All scanning activities were performed on my own system and local network in a controlled and ethical environment. No unauthorized systems were targeted.